Shreesh Yashovardhan Kolhatkar

+91 7710059933 sykolhatkar@gmail.com linkedin/shreesh-kolhatkar/ github.com/Shree21941 Computer Science undergraduate with experience in full-stack development, distributed systems, and cloud deployment. Skilled in C, Java, JavaScript, React, Docker, AWS, and algorithmic problem solving.

Education

COEP Technological University — **Aug 2024 - May 2027**
B.Tech - Computer Science and Engineering ***(CGPA: 7.61)*** — *Pune, Maharashtra*

Veermata Jijabai Technological Institute — **Oct 2021 - June 2024**
Diploma in Electronics Engineering ***(97.95%, State Rank: 15)*** — *Mumbai, Maharashtra*

Experience

Software Development Intern — **June 2025 - July 2025**
The Mother Global Foundation (TMGF) — *Pune, Maharashtra*

- Developed core features for a Volunteer Management System by building modular, responsive UI components in React.js with optimized state handling.
- Integrated REST APIs for seamless backend communication and dockerized the entire full-stack application .
- Also deployed it on an AWS EC2 instance using Nginx reverse proxy, SSL configuration, environment variables, and systemd for stable uptime.

Mechatronics & IoT Intern — **June 2023 - July 2023**
Godrej & Boyce — *Mumbai, Maharashtra*

- Developed an IoT-based water flow-rate monitoring system using Raspberry Pi, Arduino, and industrial sensors.
- Implemented real-time dashboards on Node-Red, wrote Embedded C firmware for precise sensor acquisition, and experimented with open-loop and closed-loop servo control systems.

Projects

MAC-Based SYN Flood Detection | *Computer Networks, Cybersecurity, Packet Analysis* — **April 2026**

- Implemented MAC-based filtering to improve detection reliability within local network environments compared to traditional IP-based methods.
- Utilized packet capture and real-time traffic analysis to extract features for attack detection, enabling a proactive response mechanism that reduced detection latency by 78% (13–14 ms to 3 ms).

Thermal-Aware Task Scheduler Using Distributed Systems | *MPI, Parallel Computing, Operating System* — **April 2026**

- Built a thermal-aware distributed task scheduler with a local-first execution model, reducing execution time by around 7.23% for small workloads by eliminating unnecessary cluster overhead.
- Designed a dynamic thermal migration engine that prevents CPU throttling, improving throughput by 20–23% under heavy workloads and reducing thermal-induced performance drops.

Query-Aware Context Compression for LLM Question Answering | *AI, NLP, System Optimization* — **March 2026**

- Developed a multi-stage pipeline with query decomposition, dual-stage compression (extractive + selective abstractive), and adaptive token budget allocation.
- SIntegrated retrieval, compression, and inference-time optimizations (KV cache control, sparse prefilling) into a unified system architecture.

Technical Skills

Languages & Coursework: C, Java, JavaScript, SQL, Data Structures, Algorithm Analysis, Operating System, Databses

Developer Tools: VS Code, Google Cloud Platform, Amazon Web Services, Github

Technologies/Frameworks: Linux, Docker, Kubernetes, Next.js, React.js, Express.js, MongoDB

Seminars & Sessions

CLuster Computing & MPI - COEP & C-DAC

- Delivered a session to the Non-CSE faculty about the cluster computing and MPI over physical as well as virtual machine systems

MPI Cluster and its Applicability - MIT Alandi

- Conducted a session of MPI and how the cluster computing takes place to the students of MIT-Alandi